Introduction

Standard diffusion models generate images by gradually adding Gaussian noise to training data and learning to reverse this corruption through iterative denoising. The dominant theoretical frameworks supporting diffusion such as score matching, Langevin dynamics, and variational inference all treat Gaussian noise as essential to the generative process.

Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise (Bansal et al., 2022) challenges this assumption. The authors propose that the iterative refinement loop, not the Gaussian noise, is the driving component of successful image recreation in diffusion models. They suggest that by replacing noise with arbitrary deterministic degradations such as blur and pixelation, diffusion-style generation can still produce high quality and interpretable images. Their main technical contribution is an improved sampling procedure (Algorithm 2) that is provably stable to imperfect restoration networks for linear degradations, and which empirically outperforms the naive sampler used in noise-based diffusion when applied to smooth deterministic corruptions.

Chosen Result

We aimed to reproduce the paper's central claim that Gaussian noise is not necessary for diffusion-style image restoration, specifically by reproducing the conditional reconstruction results reported in Tables 1 and 3 of the paper. We focused on two degradations: Gaussian blur and pixelation/super-resolution. We chose blur because it was the main degradation focused in the paper, and we chose pixelation because it was a non-smooth degradation that felt interesting to explore. These two degradations together test whether cold diffusion generalizes across degradation types with very different mathematical structures.

We tested on the MNIST and CIFAR-10 datasets and excluded CelebA because it directly supports the paper's main contribution while remaining feasible at our compute scale. We also designed an extension to investigate a projection-based sampling variant for pixelation to see where Algorithm 2 leaves room for improvement.

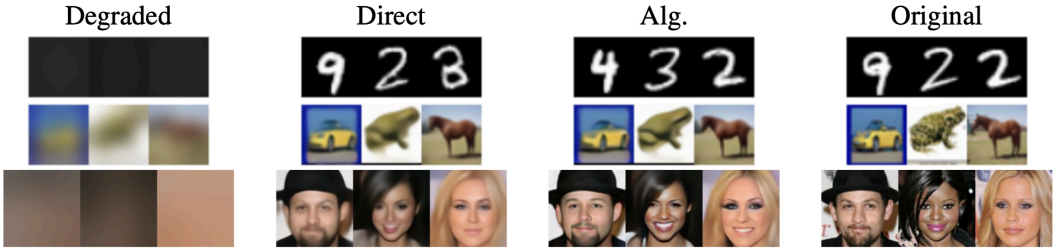

Figure 1. Blur degradation image reconstruction results (Bansal et. al)

	Degraded			Sampled			Direct		
Dataset	FID	SSIM	RMSE	FID	SSIM	RMSE	FID	SSIM	RMSE
MNIST	438.59	0.287	0.287	**4.69**	0.718	0.154	5.10	**0.757**	0.142
CIFAR-10	298.60	0.315	0.136	**80.08**	0.773	0.075	83.69	**0.775**	0.071
CelebA	382.81	0.254	0.193	**26.14**	0.568	0.093	36.37	**0.607**	0.083

Table 1. Blur degradation quantitative metrics for quality of image reconstruction (Bansal et. al)

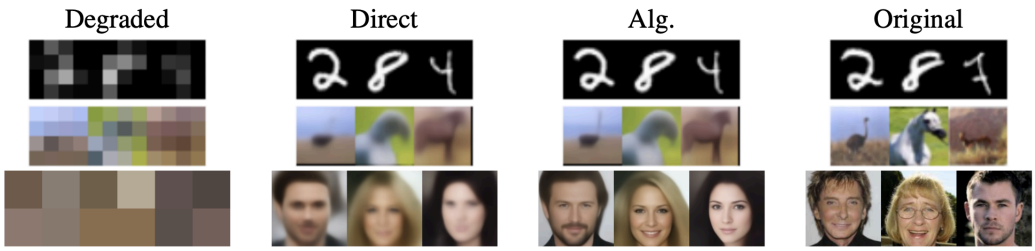

Figure 2. Pixelation degradation image reconstruction results
(Bansal et. al)

Dataset	Degraded FID	Degraded SSIM	Degraded RMSE	Sampled FID	Sampled SSIM	Sampled RMSE	Direct FID	Direct SSIM	Direct RMSE
MNIST	368.56	0.178	0.231	4.33	0.820	0.115	**4.05**	**0.823**	0.114
CIFAR-10	358.99	0.279	0.146	**152.76**	0.411	0.155	169.94	**0.420**	0.152
CelebA	349.85	0.335	0.225	**96.92**	0.381	0.201	112.84	**0.400**	0.196

Table 2. Pixel degradation quantitative metrics for quality of image reconstruction
(Bansal et. al)

Methodology

Following the paper's setup, we used MNIST (60,000 train / 10,000 test, 28×28 grayscale) and CIFAR-10 (50,000 train / 10,000 test, 32×32 RGB). For blur, we composed a sequence of Gaussian kernels with the paper's parameters (kernel size 11×11, standard deviation schedule following Bansal et al.). For pixelation, we iteratively downsampled by a factor of two per timestep, then upsampled with nearest-neighbor interpolation to maintain spatial dimensions. Each degradation produces a sequence $x_t = D(x_0, t)$ with progressively more information removed.

We trained a U-Net restoration network $R_\theta(x_t, t) \rightarrow \hat{x}_0$ that predicts the clean image from a degraded input at timestep *t*. Training minimizes the l_1 loss $||R_\theta(D(x,t),t) - x||_1$ uniformly sampled *t*, with Adam optimization at learning rate 2×10^{-5}. We trained each model for as many 350 epochs whereas the paper trained for 700k gradient steps. We evaluated direct reconstruction and iterative sampling reconstruction strategies.

Algorithm 1 Naive Sampling

Input: A degraded sample x_t
for $s = t, t-1, \ldots, 1$ **do**
  $\hat{x}_0 \leftarrow R(x_s, s)$
  $x_{s-1} = D(\hat{x}_0, s-1)$
end for
Return: x_0

Figure 3. Algorithm 1 Loop

Algorithm 2 Improved Sampling for Cold Diffusion

Input: A degraded sample x_t
for $s = t, t-1, \ldots, 1$ **do**
  $\hat{x}_0 \leftarrow R(x_s, s)$
  $x_{s-1} = x_s - D(\hat{x}_0, s) + D(\hat{x}_0, s-1)$
end for

Figure 4. Algorithm 2 Loop

Algorithm 3 Mean Sampling for Pixelation

Input: A degraded sample x_t
for $s = t, t-1, \ldots, 1$ **do**
  $\hat{x}_0 \leftarrow R(x_s, s)$
  $x_{s-1} = x_s - D(\hat{x}_0, s) + D(\hat{x}_0, s-1)$
  $x_{s-1} = \text{Project}(x_{s-1}, D(\hat{x}_0, s-1))$
end for

Figure 5. Algorithm 3 Loop

For the extension, we propose adding a projection step after each Algorithm 2 update to form Algorithm 3: $x_{s-1} \leftarrow Project(x_{s-1}, D(\hat{x}_0, s-1)$. The projection enforces that each block of x_{s-1} has its mean equal to the network's predicted low-resolution image at level $s-1$, while preserving the within-block detail produced by the network. This decouples the network's need to balance low-frequency consistency and high-frequency hallucination. The motivation is that pixelation violates Algorithm 2's smoothness assumption mentioned in the paper, so its error-cancellation property no longer holds. We believe we can take advantage of the fact that pixelation has a hard block means constraint that projection can leverage to assist the reconstruction process. To evaluate the different algorithms and each model's output, we report FID, SSIM, and RMSE computed on the held-out test sets against the original images.

Results & Analysis

Our results qualitatively reproduce the paper's main finding as both Algorithm 2 and direct reconstruction recover recognizable images from heavily degraded inputs across blur and pixelation on both datasets. Visual reconstructions match the paper's figures closely (see Figure 6). Quantitatively, our absolute FID values are poorer than the paper's (see Table 3). For example, on CIFAR-10 deblurring, we obtained sampled FID (114.36) versus the paper's (80.08), and on CIFAR-10 pixelation, we obtained sampled FID (167.13) versus the paper's (152.76). The gap is attributable primarily to reduced training duration and possible differences in FID evaluation protocol.

An unexpected finding was that in several configurations we observed that direct reconstruction outperformed Algorithm 2, which was the opposite of the paper's trend. For instance, on CIFAR-10 pixelation, our direct FID (141.15) was lower than our sampled FID (167.13). We interpret this as a limitation of the metric's ability to qualitatively assess the reproduced images. For example, in Figure 3, some sampled reconstructed MNIST images output feasible numbers, but they do not directly correspond with the numbers from the original image. This means that the metrics will be low even though the number is comprehensible.

As for the extension results, our projection-based sample produced very poor quantitative metrics. However, looking at Figure 7, the outputted images are almost interpretable. We believe that the poor results come from the fact the pixelation only uses 3 denoising steps. This means that there are only 3

iterative sampling steps as well, which means that the altered sampling algorithm is only run 3 times. We reason that perhaps the change could not provide a pronounced effect enough in 3 steps or that it was too aggressive.

The broader interpretation is that our reproduction supports the paper's central claim: Gaussian noise is not strictly necessary for diffusion-style restoration. However, our extension and our unexpected results both point to a more nuanced takeaway in that iterative refinement's value depends on degradation structure and training maturity. Algorithm 2's error-cancellation guarantee is strongest for smooth linear degradations; for non-smooth degradations like pixelation, sampler design has additional room to improve by exploiting problem-specific structure.

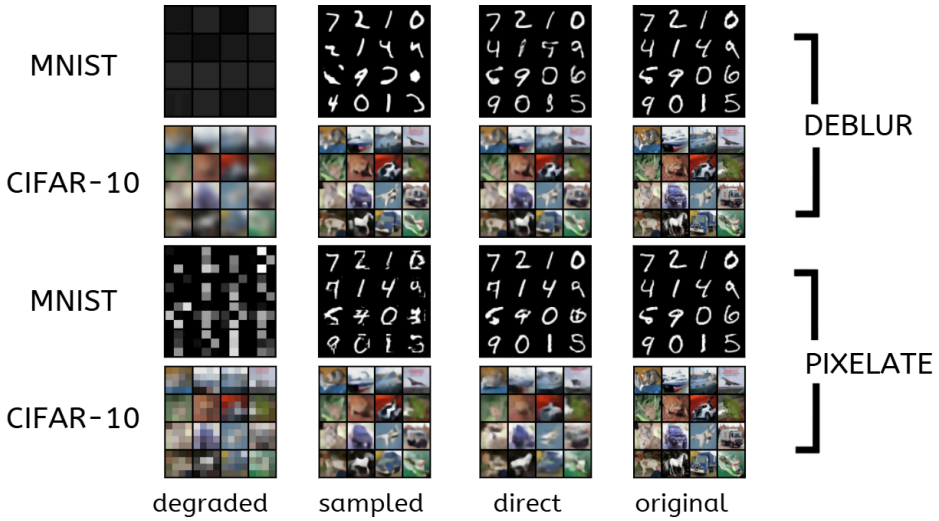

Figure 6. Reproduced results of blur and pixelation reconstruction

Dataset	Degraded			Sampled			Direct		
	FID	SSIM	RMSE	FID	SSIM	RMSE	FID	SSIM	RMSE
MNIST	406.31	0.422	0.590	66.26	0.564	0.532	27.57	0.763	0.340
MNIST*	438.59	0.287	0.287	4.69	0.718	0.154	5.10	0.757	0.142
CIFAR-10	332.07	0.290	0.269	114.36	0.707	0.150	122.21	0.724	0.140
CIFAR-10*	298.60	0.315	0.136	80.08	0.773	0.075	83.69	0.775	0.071
MNIST	303.93	0.344	0.670	48.29	0.577	0.508	19.17	0.682	0.429
MNIST*	368.56	0.178	0.231	4.33	0.820	0.115	4.05	0.823	0.114
CIFAR-10	374.33	0.250	0.230	167.13	0.330	0.273	141.15	0.325	0.277
CIFAR-10*	358.99	0.279	0.146	152.76	0.411	0.155	169.94	0.420	0.152

Table 3. Reproduced quantitative metrics; top is blur, bottom is pixelation
*from Bansal et. al

Dataset	Pixelation		
	FID	SSIM	RMSE
MNIST	180.44	0.208	0.890
CIFAR-10	144.10	0.214	0.438

Table 4. Quantitative metrics of Algorithm 3

Figure 7. Reproduced results of Algorithm 3

Reflections

Some lessons learned were that long training runs made debugging expensive since bugs in degradation operators or sampling logic could only be diagnosed after a full training cycle, which took several hours. In retrospect, we would build more lightweight sanity checks first such as training on tiny image subsets to verify the pipeline end-to-end before launching long runs, and visualizing intermediate sampling steps to catch sampler bugs without needing converged models. Keeping the codebase modular across team members was also essential, particularly as we extended the sampling algorithm.

There were several key takeaways from the paper and the reconstruction process. Firstly, the iterative refinement loop, not Gaussian noise specifically, is the core mechanism of diffusion models. Cold diffusion successfully extends the framework to deterministic degradations. Secondly, a downside of cold diffusion is that it requires degradation-specific endpoint modeling to provide a smooth distribution to draw from. This contrasts to the Gaussian noise distribution that can easily be drawn from when using hot diffusion. Lastly, iterative sampling is not automatically better than one-shot reconstruction. Its value depends on the difficulty of the prediction problem and the maturity of the restoration network.

For future directions, there are several extensions that are natural. First, full training the model to the full number of gradient steps as the paper would be very natural and helpful for the results. Second, the projection algorithm requires significant refinement. Something we could explore is adding a hyperparameter that helps tune how aggressive the added projection effect is. Along this note, the need for different endpoint modeling points to the possibility of requiring unique optimized sampling algorithms for each degradation. Last, extending cold diffusion beyond images to audio, video, or domain-specific modalities would test whether the framework's generality holds outside the vision setting.

References

1. Bansal, A., Borgnia, E., Chu, H.-M., Li, J. S., Kazemi, H., Huang, F., Goldblum, M., Geiping, J., & Goldstein, T. (2022). Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise.
2. Ho, J., Jain, A., & Abbeel, P. (2020). *Denoising Diffusion Probabilistic Models.* arXiv:2006.11239.
3. Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation.* arXiv:1505.04597.
4. LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). *Gradient-Based Learning Applied to Document Recognition.* Proc. IEEE.
5. Krizhevsky, A. (2009). *Learning Multiple Layers of Features from Tiny Images.* Technical Report.

Cold diffusion re-implementation Github repository:

https://github.com/dorazzhang/cold-diffusion